

EXAMPLE ② Solve the following ODE graphically:

$$y' = 1 + x - y$$

Solution:

STEP 1. $y' = 1 + x - y = f(x, y)$

$\Rightarrow 1 + x - y = C$ slope of the ODE

$\Rightarrow y = x + (1 - C)$

Pick values for slope, C : $C = [-1, 0, 1, 2]$

Let's find equations corresponding to $C = [-1, 0, 1, 2]$:

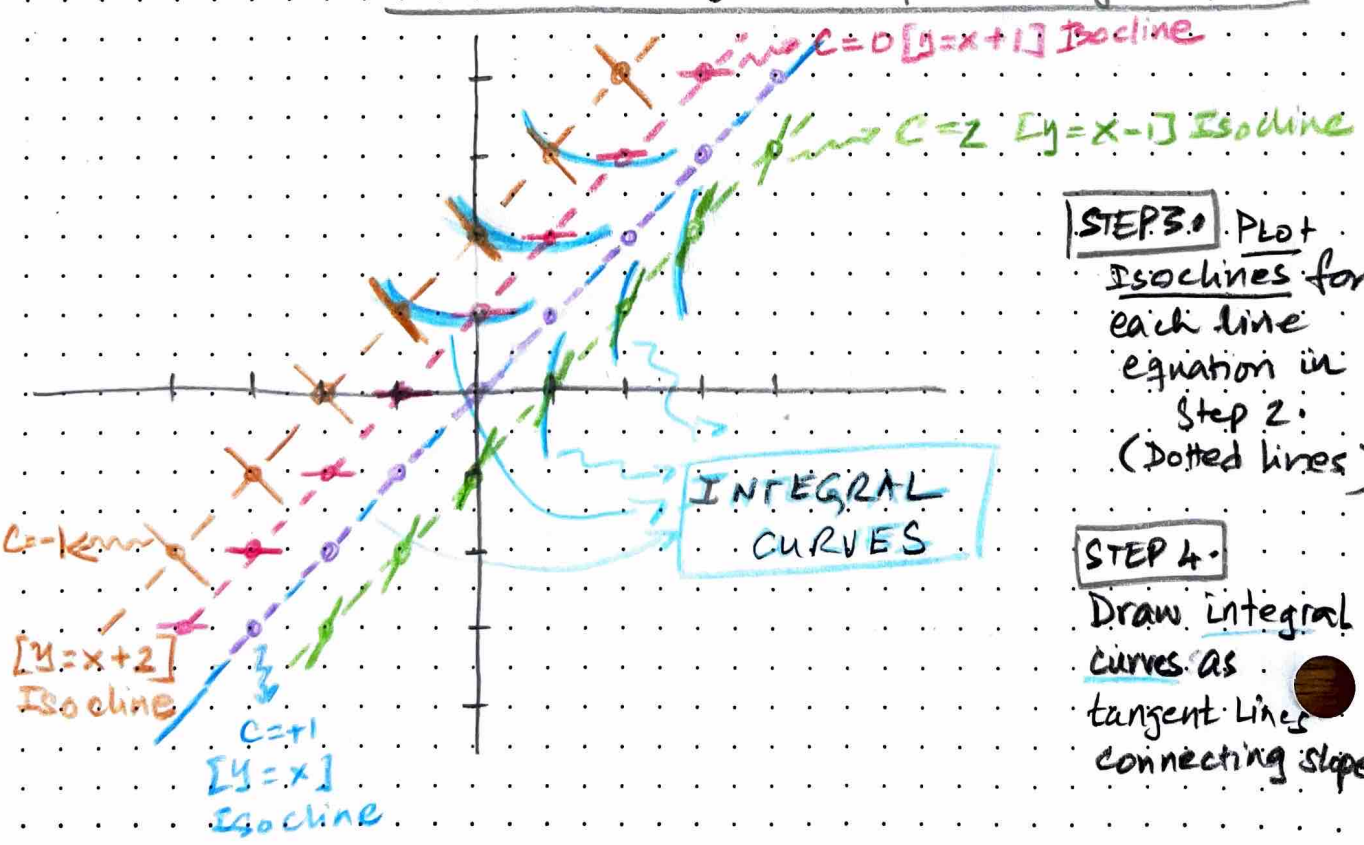
$C = -1 \Rightarrow y = x + 2$

$C = 0 \Rightarrow y = x + 1$

$C = +1 \Rightarrow y = x$

$C = 2 \Rightarrow y = x - 1$

STEP 2. Find all points (x, y) with slopes, $C = [-1, 0, 1, 2]$ and draw line segments representing, slope, C :

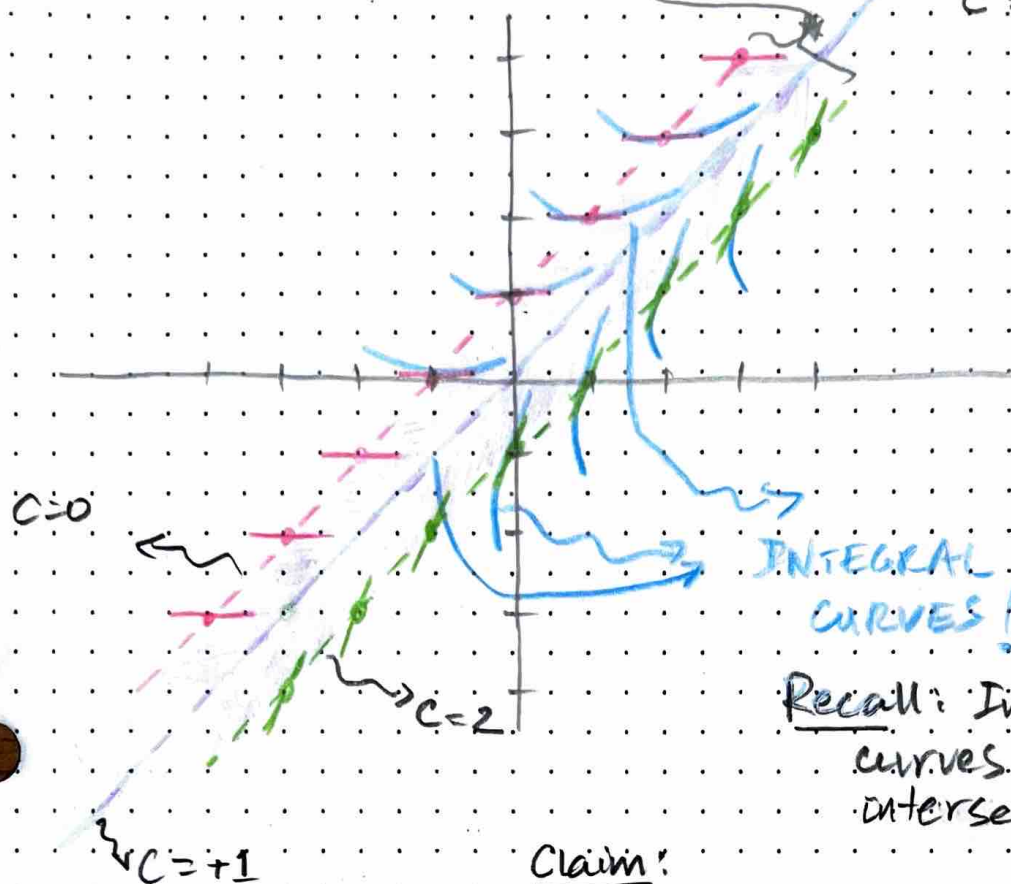


STEP 3. Plot Isoclines for each line equation in Step 2. (Dotted lines)

STEP 4. Draw Integral curves as tangent lines connecting slopes

Review w/ graphical solution:

↳ There is a corridor between isoclines $C=0$ & $C=+2$



Recall: Integral curves cannot intersect!

Claim:

↳ Isocline, $C=+1$ with equation $[y=x]$ is an integral curve.

Evaluate graphically generated integral curve $[y=x]$ using the analytical approach:

START: ODE: $y' = 1 + x - y$

- ↳ Observe on graph, $y=x$ is an integral curve
- ↳ Integral curve is a particular solution.
- ↳ $\therefore y_1(x) = x$

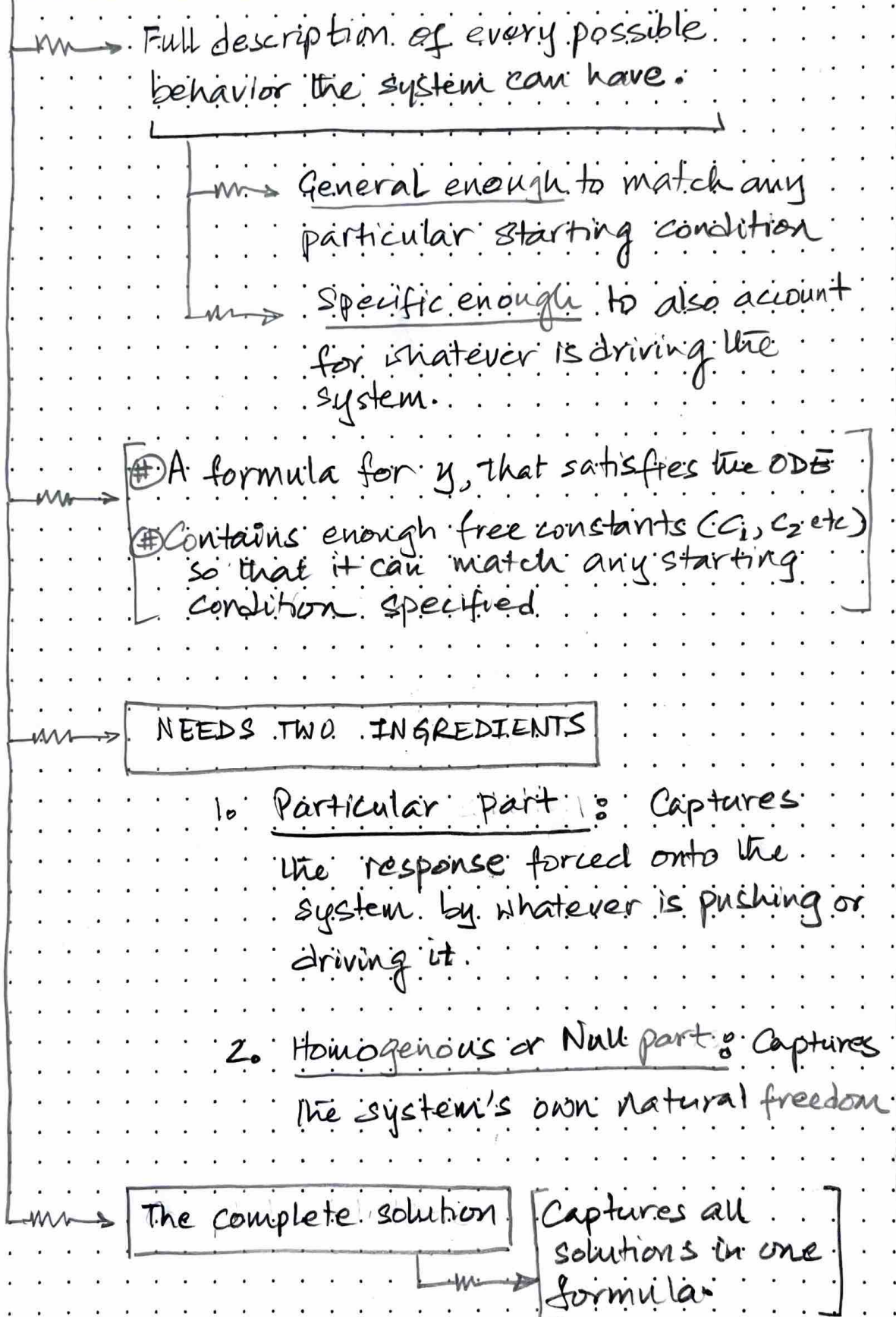
Substitute into ODE $\Rightarrow y_1' = 1 + x - y_1$

$$\Rightarrow \frac{dx}{dx} = 1 + x - x \Rightarrow 1 = 1$$

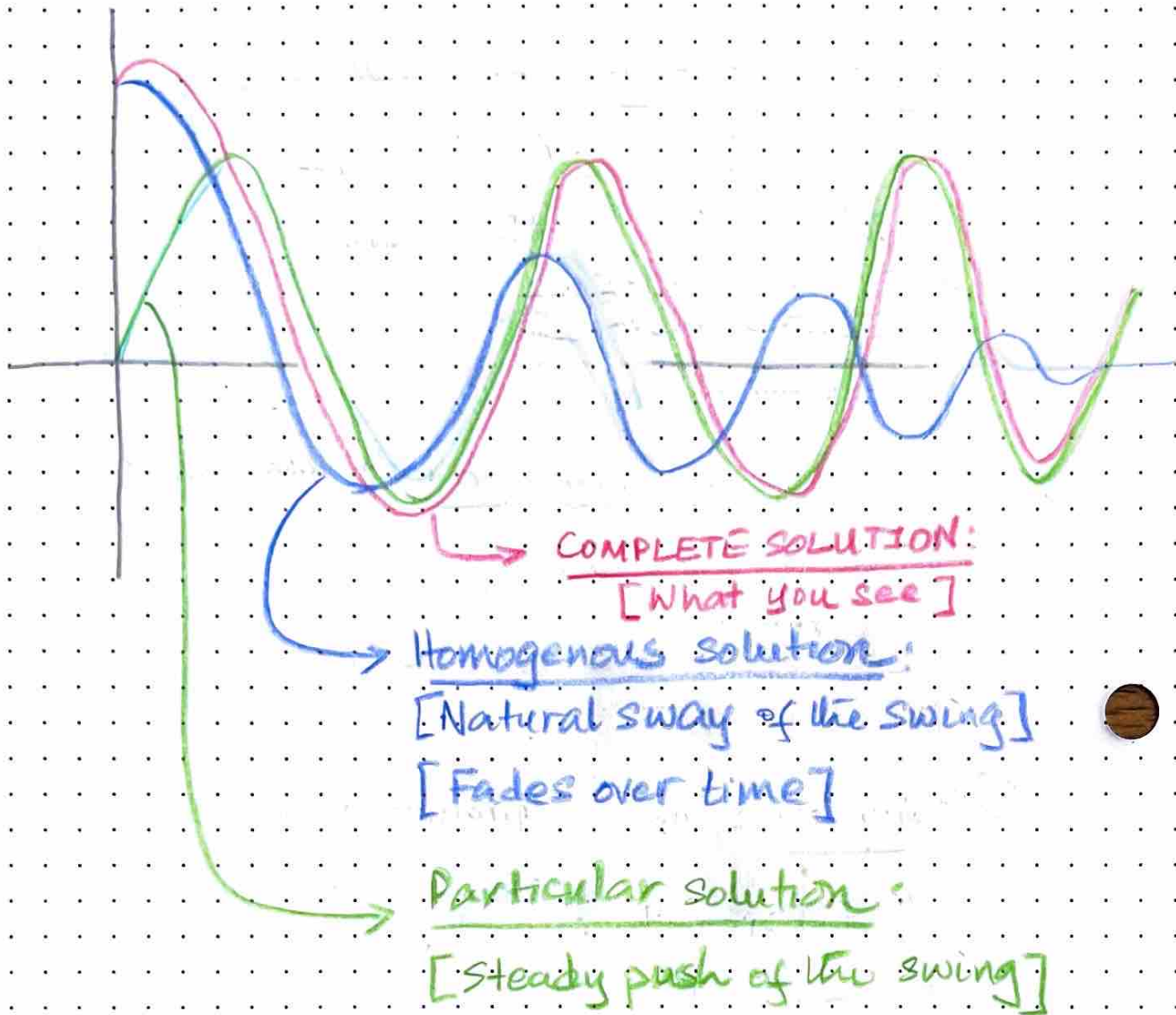
END!

$\therefore y = x$ is an (accidentally discovered) particular solution of ODE: $y' = 1 + x - y$

THE IDEA OF A

COMPLETE SOLUTION TO AN ODE

THINK OF A PLAYGROUND SWING:



Derive the complete solution for

EXAMPLE

$$y' = 1 + x - y$$

Step 1: Write the ODE in "some" standard form

$$y' = -y + (1+x) \rightsquigarrow \frac{dy}{dx} = ay + q(x)$$

① forcing term

Step 2: Find a particular solution (y_p)

2.1 We guess a particular solution of the same form as the forcing term.

$$y_p = A + Bx$$

②

Some constants

In the same form as $1+x$ term

2.2 Substitute y_p in equation ①, ODE

$$y_p' = -y_p + (1+x)$$

$$\Rightarrow \frac{d}{dx}(A+Bx) = -(A+Bx) + (1+x)$$

$$\Rightarrow B = -A - Bx + 1 + x$$

$$\Rightarrow (A+B) + Bx = 1 + x$$

2.3 Match the coefficients

$$\hookrightarrow x^1: B = 1 \rightsquigarrow ③$$

$$x^0: A+B=1 \Rightarrow A=0 \rightsquigarrow ④$$

2.4 Substitute ③ & ④ into ② $y_p = A + Bx$

Particular solution: $y_p = x \rightsquigarrow ⑤$

STEP 3:Solve the homogenous solution (y_h):(3.1) We set the forcing term to zero in the ODESubstitute y_h in the ODE

$$y_h' = -y_h + (1+x)$$

→ forcing term $q(x)$ in $y' = -y + q(x)$
 → Set $(1+x) = 0$

$$\therefore \frac{dy_h}{dx} = -y_h \rightarrow \text{This is a separable ODE}$$

$$\Rightarrow \frac{dy_h}{y_h} = -dx \Rightarrow \int \frac{dy_h}{y_h} = \int -dx + C$$

$$\Rightarrow \ln|y_h| = -x + C \Rightarrow y_h = Ce^{-x}$$

Homogenous solution

STEP 4:Combine the particular & homogenous solutions to get the complete solution (5)

$$\text{Complete solution} = \text{Eqn (5)} + \text{Eqn (6)}$$

$$y = y_p + y_h$$

$$\Rightarrow y = x + Ce^{-x}$$

complete solution

Particular part:

Captures the specific response to the forcing term, $(1+x)$

Homogenous part:

Captures general "decay" behaviour of the system

WHY THIS WORKS

→ The ODE is linear, so solutions superimpose.

$$y' = -y + (1+x)$$

$$\Rightarrow (y_p + y_h)' = -(y_p + y_h) + (1+x)$$

Rearranging terms

$$\Rightarrow (y_p' + y_p) + (y_h' + y_h) = 1+x$$

Substitute (5) & (6)

$$\Rightarrow \left(\frac{d}{dx}(x) + x \right) + \left(\frac{d}{dx} ce^{-x} + ce^{-x} \right) = 1+x$$

$$\Rightarrow (1+x) + (-ce^{-x} + ce^{-x}) = 1+x$$

$$\Rightarrow (1+x) \stackrel{!}{=} (1+x)$$

HOMOGENOUS. FIRST-ORDER ODE

(33)

KVRB

$$y' = ay \text{ with } y(0) = A$$

→ No forcing term

General solution: $\frac{dy}{y} = a dt$

Integrating $\int \frac{dy}{y} = \int a dt + C$

$$\Rightarrow \ln(y) = at + C$$

$$\Rightarrow y = Ce^{at}$$

Particular solution: $y_p(t) = Ce^{at}$

Given $y(0) = A$

$$\Rightarrow y(0) = Ce^{a(0)}$$

$$\Rightarrow C = A \quad \therefore y_p(t) = Ae^{at}$$

Complete solution = Particular solution

$$y(t) = Ae^{at}$$

EXAMPLE COMPOUNDING INTEREST

for a potential investment account:

↳ Annual compounding interest = 8%

↳ Initial investment = \$500

Solution: $a = 0.08 = 8\%$ $y(0) = 500$

$$y' = 0.08y \longrightarrow \text{1st ORDER Homogenous ODE}$$

$$\Rightarrow y(t) = Ce^{at} = Ce^{0.08t} \quad y(0) = 500 \quad \text{At } t=0$$

$$\Rightarrow y(0) = 500 = Ce^{0.08(0)} \Rightarrow C = 500$$

$$y(t) = 500e^{0.08t}$$

$$\begin{aligned} y(10) &= 500e^{0.08(10)} \\ &= \$1112.77 \end{aligned}$$

NON-HOMOGENOUS FIRST-ORDER ODE

$$\frac{dy}{dt} = ay + q(t)$$

forcing term

Case (i): $q(t) = C \rightarrow$ [constant forcing]

Case (ii): $q(t) = e^{st} \rightarrow$ [Exponential forcing]

Case (iii): $q(t) = \cos \omega t \rightarrow$ [Oscillating Input]

Case (iv): $q(t) = \text{ANY INPUT OR FORCING TERM}$

Case (v): $q(t) = \text{step or Delta function input}$

Case (vi): $q(t) = e^{i\omega t} \rightarrow$ [complex exponential forcing]

Case(ii)

RESPONSE TO EXPONENTIAL INPUT

$$\frac{dy}{dt} = ay + e^{st}$$

$y(t)$ = "Exponential response"

(Growth term) $\propto$ (Balance)

① What goes out when e^{st} goes into the ODE.

(ay)

[Interest added on to y]

(e^{st})

[Source term; Input; Deposits being made all the time.]

We have:

$\hookrightarrow$ Starting deposit

$$y = y(0) \text{ at } t = 0$$

Initial condition

e^{st}

$\hookrightarrow$ Exponentials

$\hookrightarrow$ Best possible source term for differential equations

$\hookrightarrow$ Because $\frac{d}{dt} e^{st} = e^{st}$

$\hookrightarrow$ Easiest to work with

SOLUTION :

$$\text{Full solution} = [\text{Particular solution}] + [\text{Homogenous (or Null) solution}]$$

Particular solution (y_p):

$\hookrightarrow$ Since we have e^{st} , as a "nice" source function look for multiples of e^{st} as a particular solution.

Particular solution (y_p):

- Handles the "source or forcing" term
- Or the non-zero input function
- Provides one specific function that satisfies the differential equation
- Contains no arbitrary constants

③ $y_p = Y e^{st}$ → Find: Y (is a number)
 And I have a solution (particular)
 for $\frac{dy}{dt} = ay + e^{st}$

Substitute $y_p = Y e^{st}$ into DDE, and
 Solve for Y .

④ $\frac{dy_p}{dt} = ay_p + e^{st}$ → for particular solution (y_p):
 Substitute y_p in ①

$$\Rightarrow \frac{d}{dt}(Y e^{st}) = a Y e^{st} + e^{st}$$

$$\Rightarrow Y s e^{st} = a Y e^{st} + e^{st}$$

$$\Rightarrow (s - a) Y = 1 \quad \therefore Y = \frac{1}{s - a}$$

From ③ & ⑤

⑥ $y_p = \left(\frac{1}{s - a} \right) e^{st}$

$$y_p = \left(\frac{1}{s-a} \right) e^{st}$$

→ Particular solution

→ One part of the full solution.

→ Will not match the initial condition

[How can I match the initial condition?]

$$y = y(0) \text{ @ } t=0$$



→ I need the null solution or homogenous solution.



Recall: Full solution = $\overset{(y_p)}{\text{Particular solution}} + \overset{(y_h)}{\text{Homogenous solution}}$

Homogenous solution (y_h):

- Handles the internal dynamics of the system
- when the "forcing or input" term is set to zero
- Contains arbitrary constants to allow the solution to adapt to any starting conditions

$$\boxed{\frac{d}{dt} y_h = a y_h + e^{st}} \quad \leftarrow \text{For homogenous solution:}$$

1. Substitute y_h in (1)
2. Set $e^{st} = 0$

(7)

$$\Rightarrow \frac{dy_h}{dt} = a y_h \Rightarrow \frac{dy_h}{y_h} = a dt$$

Integrating on both sides

$$\int \frac{dy_h}{y_h} = \int a dt + C_1$$

$$\Rightarrow \ln y_h = at + C_1$$

$$\Rightarrow y_h = e^{C_1 + at} = e^{C_1} e^{at}$$

$$\therefore \boxed{y_h = C e^{at}} \quad \rightarrow (8)$$

Full solution:

Recall: $\boxed{y(t) = y_p(t) + y_h(t)} \quad \rightarrow (9)$

Full Solution
Particular Solution
Homogenous Solution

From (6), (8) & (9)

$$\boxed{y(t) = \frac{e^{st}}{s-a} + C e^{at}} \quad \rightarrow (10)$$

Verify initial condition, equation (2)
(or plug in) $\hookrightarrow$ to find C $[y = y(0) @ t = 0]$

$$y = y(0) \text{ @ } t = 0 \rightarrow \text{Eqn. (2) Recall}$$

Substitute equation (2) in (10) $\rightarrow$ full solution

$$y(t) = \frac{e^{st}}{s-a} + C e^{at}$$

$$\Rightarrow y(0) = \frac{e^{s(0)}}{s-a} + C e^{a(0)}$$

$$\Rightarrow y(0) = \frac{1}{s-a} + C \quad \therefore C = y(0) - \left(\frac{1}{s-a}\right)$$

(11)

Full (or complete) solution now satisfies initial

condition

From (10) & (11)

$$y(t) = \left(\frac{e^{st}}{s-a}\right) + \left[y(0) - \left(\frac{1}{s-a}\right)\right] e^{at}$$

Re-arranging the terms in the equation above,

$$y(t) = \left(\frac{e^{st} - e^{at}}{s-a}\right) + y(0) e^{at}$$

(12)

Particular Solution:

⊕ Exponential source or input grows this way

Part that is growing from the source term

"Additional deposits" (e^{st}) that are contributing to growth

Part that is growing from the initial condition

"Initial deposit" that is growing

Null or Homogeneous Solution:

⊕ Initial condition grows this way

Just when we thought the problem is solved

...



Let's take a closer look at the full solution

$$y(t) = \left(\frac{e^{st} - e^{at}}{s-a} \right) + y(0)e^{at}$$

→ This form of the full solution has an exception!



The solution "blows up" or "breaks down" at $s=a$!



At $s=a$, the ODE describes a system experiencing RESONANCE.

Notice at $s=a$: Equation (12)

$$y(t) = \frac{e^{at} - e^{at}}{a-a} + y(0)e^{at}$$

(13) $\left(\frac{0}{0} \right)$ Indeterminate!

Resonance:

→ In an ODE occurs when the forcing term is matching the natural growth of the homogeneous system

[Causes the system to grow unboundedly over time]

We need to deal with "Resonance"

→ arising due to $\frac{0}{0}$ or $\frac{\infty}{\infty}$

→ We use: L'Hôpital Rule

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

→ Apply L'Hôpital Rule to the term $\left(\frac{e^{st} - e^{at}}{s - a} \right)$ in eqn (12).

$$\Rightarrow \lim_{s \rightarrow a} \left(\frac{e^{st} - e^{at}}{s - a} \right) = \lim_{s \rightarrow a} \frac{\frac{d}{ds}(e^{st} - e^{at})}{\frac{d}{ds}(s - a)}$$

$$= \lim_{s \rightarrow a} \left(\frac{te^{st}}{1} \right) = te^{at}$$

$$\Rightarrow \left(\frac{e^{st} - e^{at}}{s - a} \right) = te^{at}$$

Substituting (14) in (12) we have the full solution:

$$y(t) = te^{at} + y(0)e^{at}$$

Full solution now satisfies
 (1) Initial condition, $y(0)$
 and
 (2) Resonance

Note: Expect a term with factor, t , when there is resonance.